

DAT535 – Data Engineering Fall 2026



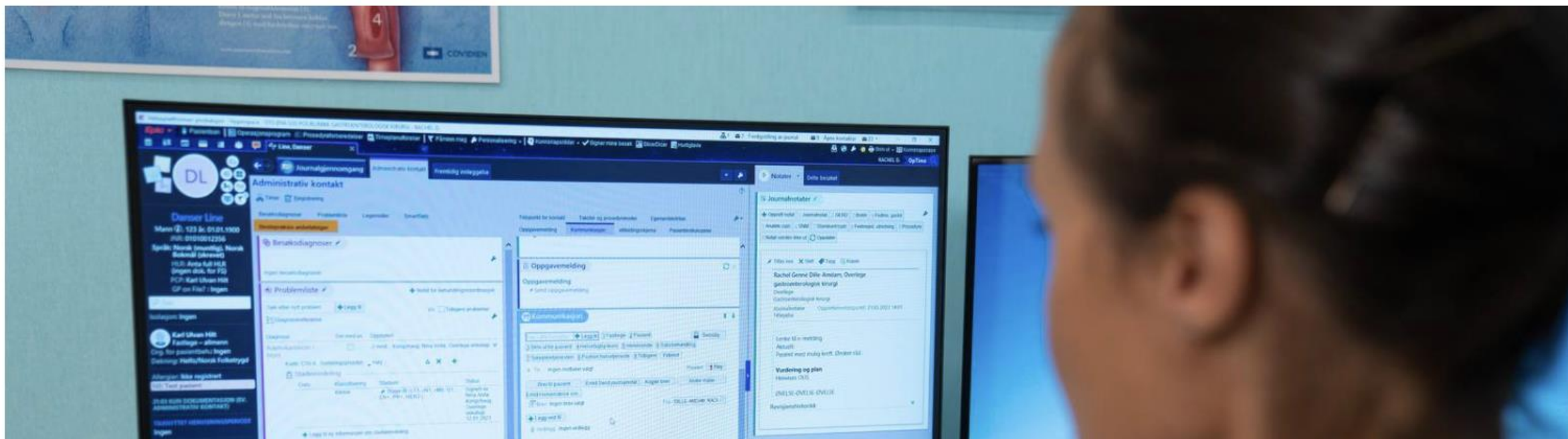
University
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Data Platforms

Key concepts

Helseplattforma får leve vidare

Ingen fekk fleirtal for å skrote Helseplattforma då Stortinget stemte i dag. Dermed held det omstridde journalsystemet fram – trass i sterk kritikk frå både politikarar og brukarar.



Main types of platforms

- Databases
- Data warehouses
- Data Lakes/Lakehouses/Fabric
- Data Mesh

Databases: reliable operational truth

The relational model made application data easier to structure, connect and protect.



1970: relational model

Rows + columns + declared relationships

What problem did it solve?

Run business transactions correctly and repeatedly.

- 1 **TABLES** Structure data as rows and columns
- 2 **KEYS** Give each record an identity
- 3 **RELATIONSHIPS** Connect records without repeating everything
- 4 **NORMALIZATION** Reduce duplication and update anomalies
- 5 **INTEGRITY** Use constraints and transactions to keep data valid

LIMIT

**Optimized for transactions,
not cross-system analytics.**

If we had SAP, Salesforce and Timestream data, where would you put them to create one management report?

Data warehouses: one consistent view for analysis

Separate analytical work from operational systems, then model data around business questions.



1980s–1990s: analytical data

Sources → ETL → warehouse → reporting

What problem did it solve?

Combine historical data into a trusted analytical view.

- | | | |
|---|--------------|--|
| 1 | OLTP vs OLAP | Transactions optimized for change; analytics optimized for reading |
| 2 | FACT TABLES | Measurements at a defined business process and grain |
| 3 | DIMENSIONS | Business context such as time, asset, product or customer |
| 4 | STAR SCHEMA | Simple fact-centered model for understandable queries |
| 5 | SNOWFLAKE | Normalized dimensions with more structure and joins |

LIMIT

Excellent for known reporting needs; less flexible for new data types.

What if we suddenly want to use PDFs, images, sensor data, logs, or ML?

Lake → Lakehouse → Fabric: from storage to platform

Keep diverse data cheaply, add reliable table behavior, then unify the tools that use it.



2010s → 2020s: unify data workloads

Lake storage → reliable tables → shared services

What problem did it solve?

Support analytics and AI across more data types and workloads.

- 1 DATA LAKE** Object storage for structured, semi-structured and unstructured data
- 2 OPEN TABLES** Parquet plus Delta or Iceberg for interoperable, reliable tables
- 3 ELT + PROCESSING** Land first; transform with batch, streaming, SQL or Spark
- 4 MEDALLION** Bronze raw → Silver refined → Gold business-ready
- 5 FABRIC** OneLake with integrated engineering, warehouse, BI and AI tools

LIMIT

A unified platform does not define business meaning or ownership.

If every domain can publish data into Fabric, who is responsible when a production dashboard shows wrong numbers?

Data Mesh: scale ownership, not only technology

Move analytical data responsibility closer to the domains, supported by a shared platform and common rules.



2019 onward: sociotechnical design

Domain products + shared platform + federated rules

What problem did it address?

Central data teams become a knowledge and delivery bottleneck at scale.

- 1 DOMAIN OWNERSHIP** Domains carry responsibility for their analytical data
- 2 DATA AS PRODUCT** Published data is usable, described, governed and maintained
- 3 SELF-SERVE PLATFORM** Common capabilities let domains deliver without bespoke support
- 4 FEDERATED GOVERNANCE** Shared policies preserve security and interoperability
- 5 SOCIO-TECHNICAL** Team boundaries, incentives and architecture must fit together

LIMIT

Decentralization without capability and standards creates more silos.

Discussion

1. Why can't everything stay in databases?
2. If storage is cheap, why not keep everything forever?
3. Who should own a number in the dashboard?

Summary

LAYER	MAIN QUESTION	KEY CONCEPTS
Database	How do I store and update data correctly?	Relational model, Tables, Keys, Relationships, Normalization, ACID
Data Warehouse	How do I analyze business performance?	OLTP vs OLAP, Fact Tables, Dimensions, Star Schema, KPI Modeling
Lake / Lakehouse / Fabric	How do I manage data at scale for analytics, AI, and operations?	Raw-to-curated pipelines, Medallion Architecture, Open Formats, ELT, Unified Analytics
Data Mesh	How do I organize ownership and governance across domains?	Domain Ownership, Data Products, Self-Service Platform, Federated Governance